Intelligent Predictive Decision Support System for Optimized Multi-Horizon Electric Vehicle Charging in Smart Cities

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***Abstract*—The rapid growth of electric vehicles (EVs) in smart cities creates issues for controlling charging demand and power grid reliability. In this work, we conduct a systematic analysis of 15 research papers to investigate how EV charging demand may be forecasted across short, medium, and long time horizons, and how these predictions can be used to optimize charging for both customers and the power grid. While powerful machine learning and forecasting models can accurately predict charging demand, the majority of present research approaches forecasting and decision-making as different concepts. As a result, forecasts are rarely translated into useful charging recommendations.This analysis highlights the importance of integrated multi-horizon forecasting mixed with intelligent decision support systems, and it lays the groundwork for future research on optimum EV charging in smart cities.**

***Index Terms*—Electric vehicle charging, smart cities, load forecasting, multi-horizon prediction, decision support systems, grid-aware charging, charging optimization.**

I. Introduction

In this report, we perform a systematic literature review (SLR)to study how electric vehicle (EV) charging demand may be forecasted across multiple time periods in smart cities, and how these predictions can be utilized to optimize charging for both EV users and the power grid. We chose this issue because EV adoption is quickly expanding in urban areas, driven by environmental aims, emission reduction targets, and smart city programs.

While EVs have environmental benefits, their charging behavior presents significant hurdles to urban power infrastructure, according to our preliminary research. EV charging demand is highly variable and driven by a number of factors, including user behavior, traffic patterns, electricity cost, renewable energy availability, and grid constraints. Uncontrolled charging can cause peak load concerns, voltage instability, congestion at charging stations, and lower user satisfaction.

One key finding from the research is that EV charging demand varies across time horizons. Short-term demand influences real-time grid operation and charging station control; medium-term demand facilitates operational planning; and long-term demand is critical for infrastructure development and policy considerations. However, many previous research focus on a single time horizon at a time.

In addition, we discovered that prediction alone is insufficient. Grid operators and community managers require decision support systems (DSS) that can convert forecasts into practical charging schemes. Although the rising amount of research on EV charging prediction, the combination of multi-horizon forecasting and intelligent decision support is still limited.

Based on these observations, we designed this study around the following research questions:

- RQ1:How can EV charging demand be accurately predicted across different time periods in smart cities?
- RQ2: How might these forecasts be used to optimize charging for customers and the power grid?

II. BACKGROUND AND RELATED WORK

A. Background

The rapid growth of electric vehicles (EVs) is a key component of smart city development, driven by the need to reduce carbon emissions and increase urban sustainability. As the number of electric vehicles grows, rises the demand for charging infrastructure and electricity. Unlike traditional electrical loads, EV charging demand is very unpredictable and is determined by factors such as user behavior, schedules, electricity pricing, and the availability of charging stations. If charging is not effectively managed, it can result in grid congestion, higher electricity bills, long wait times at charging stations, and lower consumer satisfaction.

To solve these difficulties, accurate forecasting of EV charging demand is critical. Forecasting enables grid operators and city planners to forecast demand effectively manage resources over many time horizons. Short-term forecasting enables real-time grid operation, but medium- and long-term forecasting aids in operational planning and infrastructure development. However, forecasting alone is not enough. Smart cities also require decision support systems that can utilize these forecasts to recommend the best charging procedures for both EV users and the power grid.

B. Related Work

Previous research on EV charging in smart cities [1] has primarily focused on establishing models for charging demand prediction. Many studies employ machine learning and deep learning techniques to predict EV charging demand over short, medium, and long time periods [2], while others use probabilistic approaches to account for uncertainty in user behavior. Other works use contextual data including time, place, and mobility patterns to improve prediction accuracy. These studies show that modern data-driven systems can accurately anticipate EV charging demand.

In addition to forecasting, several studies look on grid-aware charging optimization and decision support systems [3]. These approaches take into account grid constraints, electricity costs, and charging station capacity in order to reduce peak loads and increase system stability [4]. However, most existing solutions treat forecasting, optimization, and decision support individually, and they frequently focus on a single time horizon [5]. Integrated frameworks that combine multi-horizon forecasting [6]with intelligent decision assistance for both users and the power grid are rare, indicating a significant gap in current research.

III. RESEARCH QUESTIONS

RQ1: How can EV charging demand be accurately predicted across different time periods in smart cities?
Objective: To study and compare existing forecasting techniques for predicting EV charging demand in smart cities over short, medium, and long time periods.

RQ2: How might these forecasts be used to optimize charging for customers and the power grid?
Objective: To investigate how EV charging demand estimates are used in optimization and decision-support systems to improve charging efficiency, user convenience, and power grid reliability.

IV. RESEARCH METHODOLOGY

To answer our research questions, we used a systematic literature review methodology that ensured transparency and relevance throughout the review process.

A. Literature Search Process

We performed our literature search utilizing academic databases such as IEEE Xplore and Google Scholar. We selected independently reviewed journal publications and high-quality conference papers on EV charging, smart cities, forecasting, optimization, and decision support.

B. Keyword and Search Strategy

We utilized combinations of these keywords:

- Forecasting electric vehicle charging demand.
- Short-term, medium-term, Long-term and load forecasting.
- Smart city energy management.
- Grid-aware EV charging, decision support systems.
- Charging optimization.
- Scheduling.

As we analyzed relevant articles, we revised our keywords to better incorporate research on prediction and decision-making.

C. Inclusion and Exclusion Criteria

To ensure the significance and standards of the selected studies, we used defined inclusion and exclusion criteria. Studies were included if they focused on EV charging demand prediction, optimization, or decision support, particularly in smart city or urban power grid contexts. Papers on short-, medium-, and long-term forecasting, as well as the application of forecasts to improve charging efficiency and grid performance, were considered significant.

Studies were eliminated if they did not directly address EV charging or its influence on the power system. Papers that concentrated primarily on car hardware, battery technology, or associated power system subjects were not evaluated. Anonymous materials, such as reports, blogs, or opinion articles, were also removed to ensure the accuracy of scientific research.

D. Selection Criteria:

The papers were selected based on:

- Relevance to EV charging demand forecast or optimization.
- Impact to smart cities or urban power grids.

- Utilization of data-driven, probabilistic, or optimization-based methods.
- Reviewed publication quality.

After examination, we picked 15 research papers that provide a comprehensive picture of forecasting, uncertainty management, grid-aware charging, and decision assistance.

E. Data Extraction and Analysis

We gathered important information from each study that was relevant to our research questions. This covered the forecasting horizon (short, medium, or long term), modeling or optimization methodologies employed, data type, and how forecasting results were used for charging optimization or decision assistance.

The gathered data was then evaluated and compared across studies. Rather than describing each work separately, we organized them according to topic to find shared methods, variances, and limits. This gave us insight into how EV charging demand prediction is currently handled and how forecasts are applied in practice.

F. Synthesis of Findings

Finally, we combined the selected research' findings by combining forecasting techniques with optimization and decision support strategies. This synthesis revealed patterns, strengths, and gaps in existing research, particularly in the absence of integrated multi-horizon forecasting and intelligent decision support systems.

This organized synthesis served as the foundation for answering our study objectives and determining future directions for research on optimum EV charging in smart cities.

V. RESULTS

RQ1: Predicting EV Charging Demand Across Different Time Periods

Short-Term Forecasting

Based on the studied literature, we determined that short-term forecasting (minutes to hours ahead) is mostly employed for real-time grid operation and charging station control [2]. Many studies use machine learning and deep learning techniques as LSTM, Seq2Seq, Transformer models, and ensemble learning. These methods are efficient for capturing temporal pricing patterns and rapid demand shifts.

However, we discovered that most short-term forecasting research consider prediction as a stand-alone problem. While they achieve great accuracy, they frequently fail to explain how forecasts are applied in real-world billing decisions or grid control. [7]

Medium and Long-Term Forecasting

Medium- and long-term forecasts are widely utilized for infrastructure development, capacity growth, and regulation. [8] Neural networks, radial basis function (RBF) models, and hybrid forecasting methodologies are commonly used. These studies highlight the significance of seasonal cycles, economic considerations, and EV adoption growth. [2]

Despite their value, we discovered that long-term forecasting models are frequently separated from short-term operational requirements. They are rarely integrated into systems that provide real-time or immediate decision-making. [7]

Probabilistic and Uncertainty-Aware Approaches:

Several research use probabilistic forecasting to handle the uncertainty in EV charging behavior [1]. These solutions acknowledge that user behavior and environmental factors are not deterministic. Probabilistic models create more solid estimates by taking into account numerous demand scenarios.

However, we discovered these approaches are frequently computationally expensive and primarily evaluated using simulations, restricting their use in real-world operational systems.

RQ2: Using Forecasts to Optimize Charging for Customers and the Grid

Grid-Aware and Context-Aware Charge Optimization

Our analysis shows that grid-aware charging solutions are gradually incorporating grid constraints [3]such as transformer capacity, voltage limits, and power price. Optimization techniques and model predictive control (MPC) are utilized to coordinate charging with grid conditions, reducing peak loads and enhancing stability. [4]

These approaches provide clear benefits for grid operations. However, we discovered that many studies focus on specialized use cases, such as individual charging stations or public transportation vehicles, rather than city-wide charging infrastructures [5].

Decision Support Systems for Electric Vehicle Charging

Only a few research specifically recommend decision support tools for EV charging. Existing DSS solutions [9] are often focused on a specific goal, such as reducing costs or emissions, and operate over a single time period. [1]

We discovered a clear gap in the literature: forecasting results are rarely translated into user- or grid-oriented recommendations, such as optimal charging times or locations. As a result, customers and grid operators do not get the full benefit of accurate predictions [4].

VI. Discussion

Based on the results of our evaluation, we have identified three main gaps in current research.

First, multi-horizon integration is lacking. While there are accurate models for short, medium, and long-term forecasting, they are rarely merged into a coherent framework. In true smart cities, decisions made currently have long-term consequences.

Second, decision support is insufficient. Many studies finish at prediction or optimization without making practical suggestions to stakeholders. Effective EV charging management requires systems that link forecasts and choices.

Third, real-world complexity is frequently simplified. User behavior, traffic, renewable energy availability, price of electricity variations, and grid limits are rarely considered simultaneously.

VII. CONCLUSION

In this study, we conducted a systematic review of 15 research papers to answer two research questions about EV charging demand forecast and optimization in smart cities. We discovered that, while accurate forecasting methods exist for various time periods, present research lacks integrated systems that employ these projections to assist informed charge decisions.

This review provides a solid framework for our future master's thesis work, which will be focused on creating and assessing an intelligent decision support system that benefits both EV users and the power grid by enabling efficient, grid-aware, and user-centric charging approaches.

VIII. Division of Work

The task was divided equally among the two group members:

Literature Search: We both (Joshna and Pujitha) conducted a combined search and selection of relevant research papers.

Analysis: Pujitha concentrated on forecasting and uncertainty-aware methods, while the other Joshna analyzed optimization and decision support techniques.

WRITING: All sections were written together, with shared modifications to guarantee clarity and consistency.

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